AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 3 · STUDENT LESSON

Branching Logic

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

AI-1 Week 10: distinguish strict and inclusive conditions.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Trace the selected branch in an if elif else chain. Try the recall before opening notes.

Your learning target

I can trace the selected **branch** in an if elif else chain.

I can use **indentation** to identify which statements belong to a **branch**.

I can test values at and around a decision **boundary**.

Remember before reading

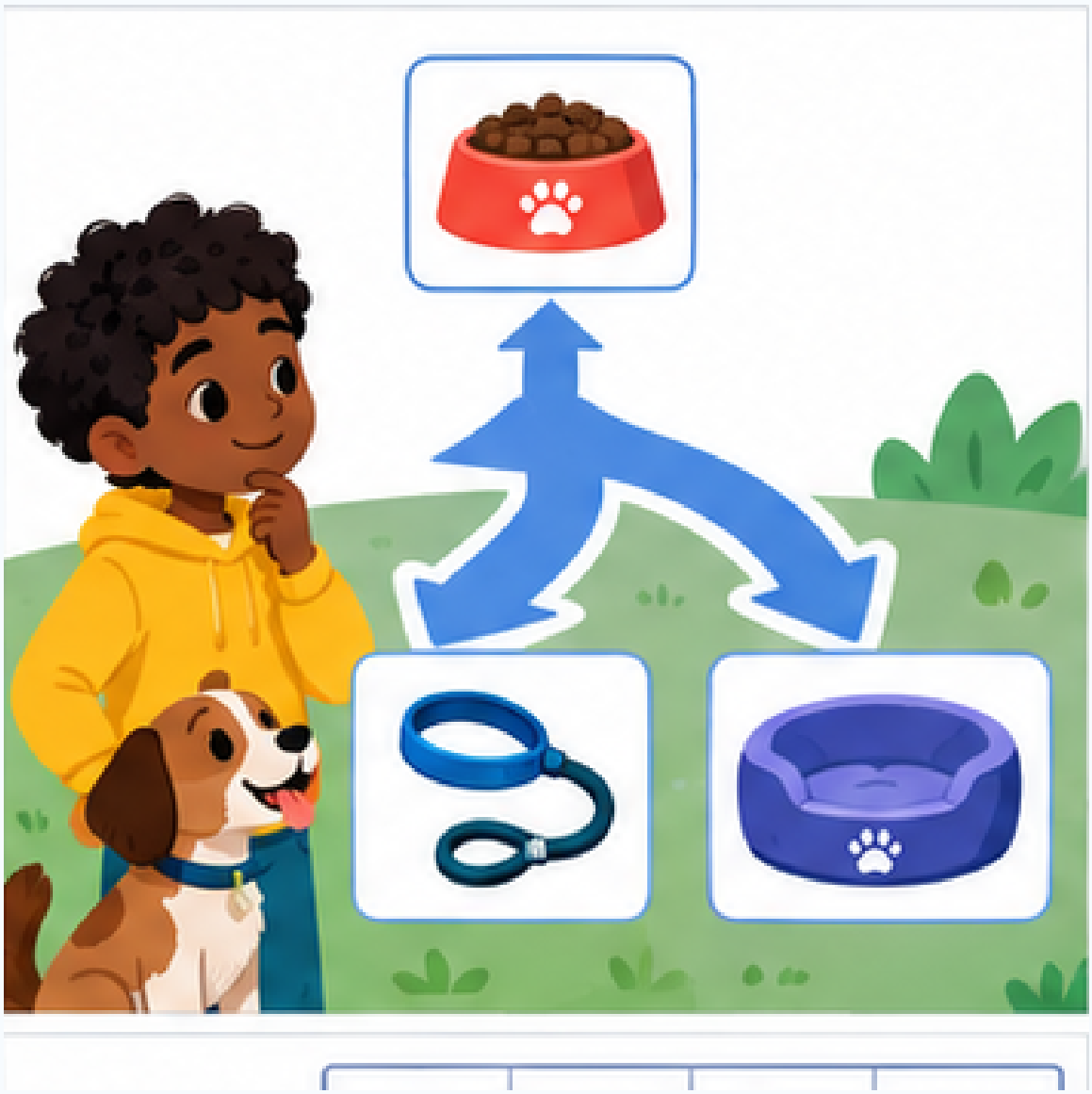
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 3: **Branch**ing Logic

Ordered **condition**s determine which path a system takes.



THINKER MISSION

Build and trace a three-**branch** pet-care algorithm.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Does minutes <= 12 accept exactly 12? How does < differ?

This week's end goal

By the end, I can **order branch**es so each **input** receives exactly its intended category.

STUDENT LESSON · 2 OF 18

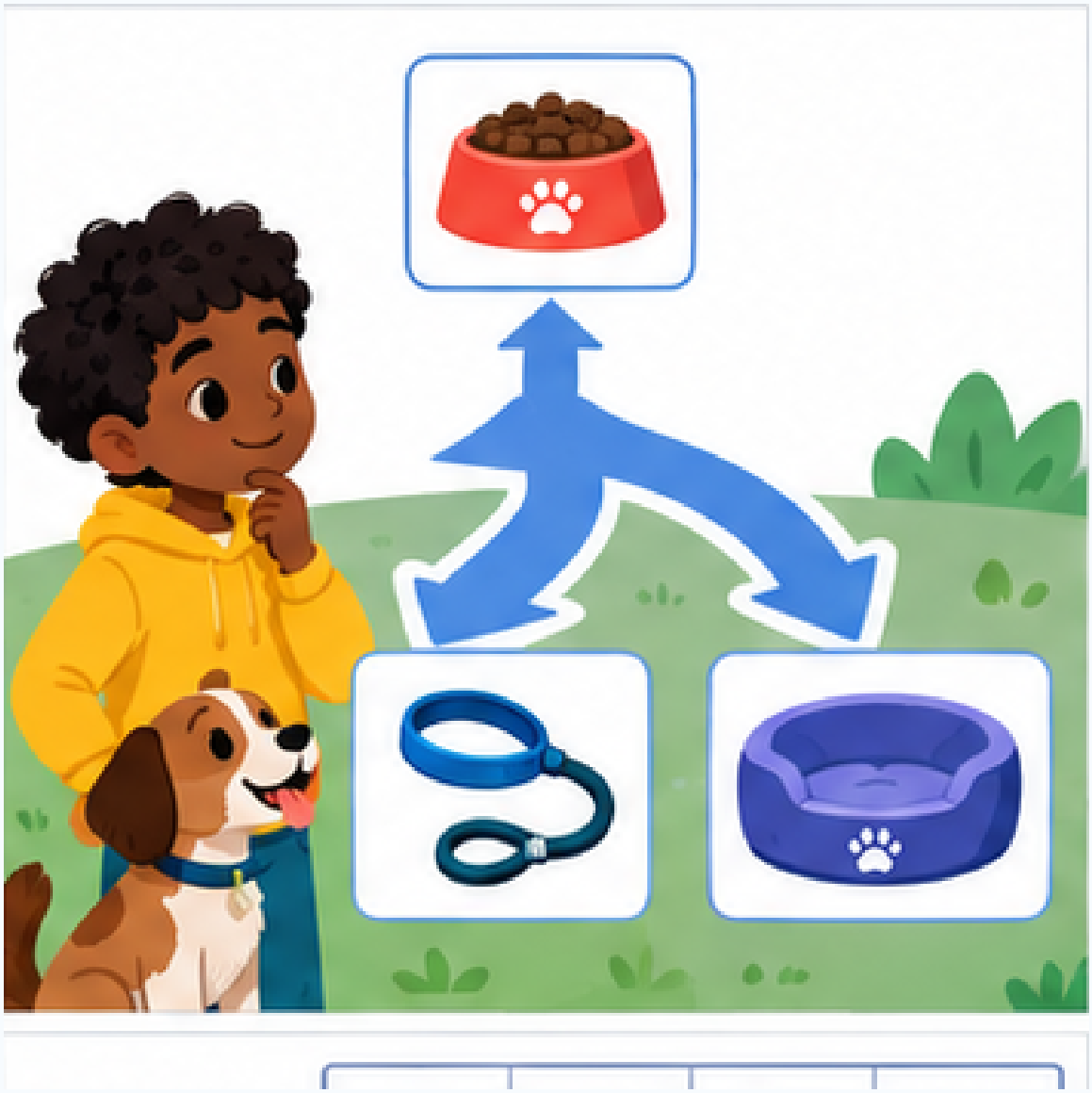
Notice the evidence

Investigate the supplied Branching Logic case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **branch**ing logic to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional activity sorter labels a supplied integer duration. Negative values are Invalid; values from zero through 12 are Short; larger values are Long. The labels describe duration only.

Write this requirement before the program so that a test has an expected answer independent of the code.

An if statement selects a block when its **condition** is true. An elif is considered only if earlier **condition**s in the same chain were false. An else handles the remaining cases. Colons end the **condition** lines.

Use four spaces for the indented statements and keep related **branch**es aligned.

STUDENT LESSON · 3 OF 18

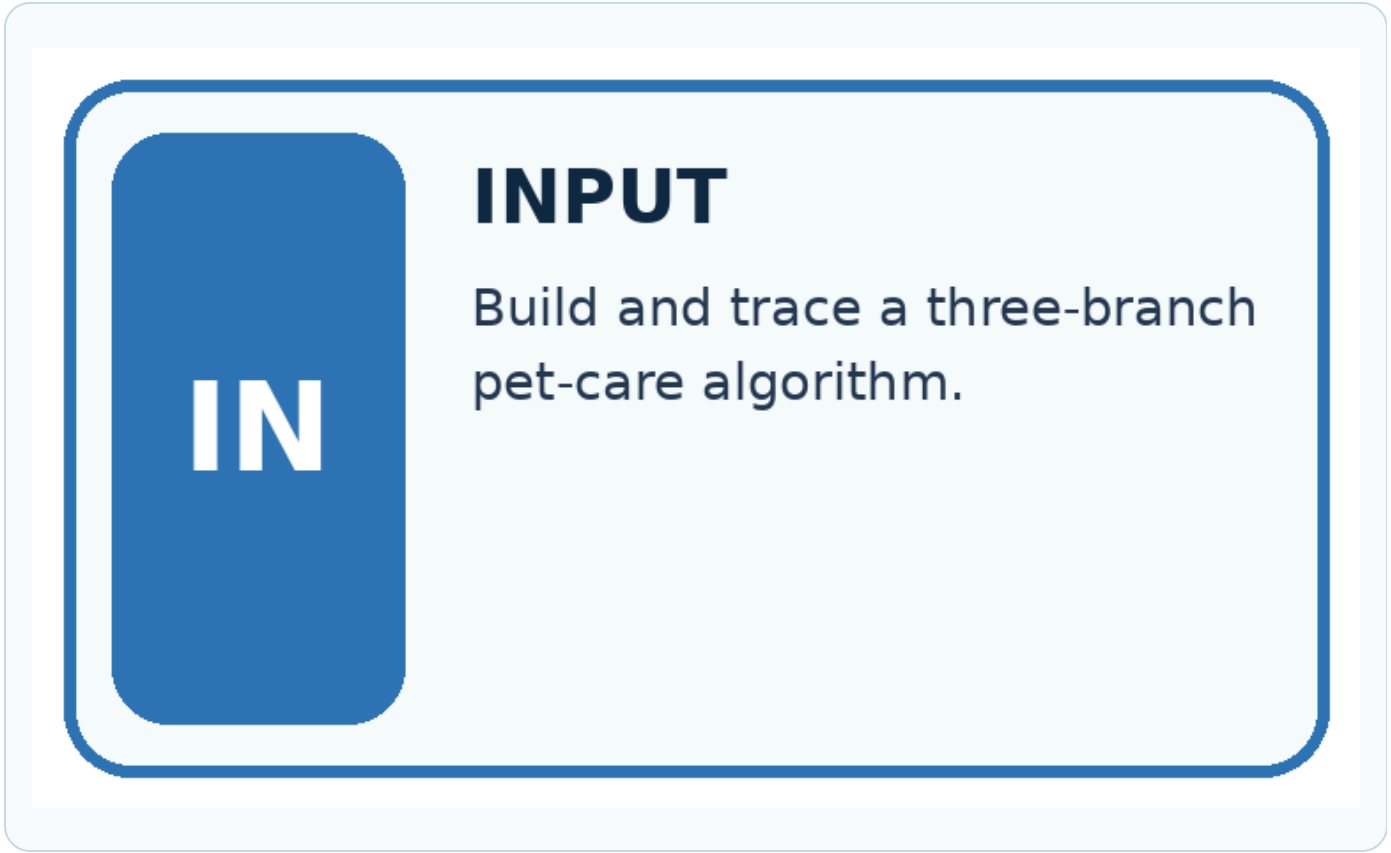
See how it works

> Follow the mechanism for Branching Logic. Connect each labelled input, deciding step and result.

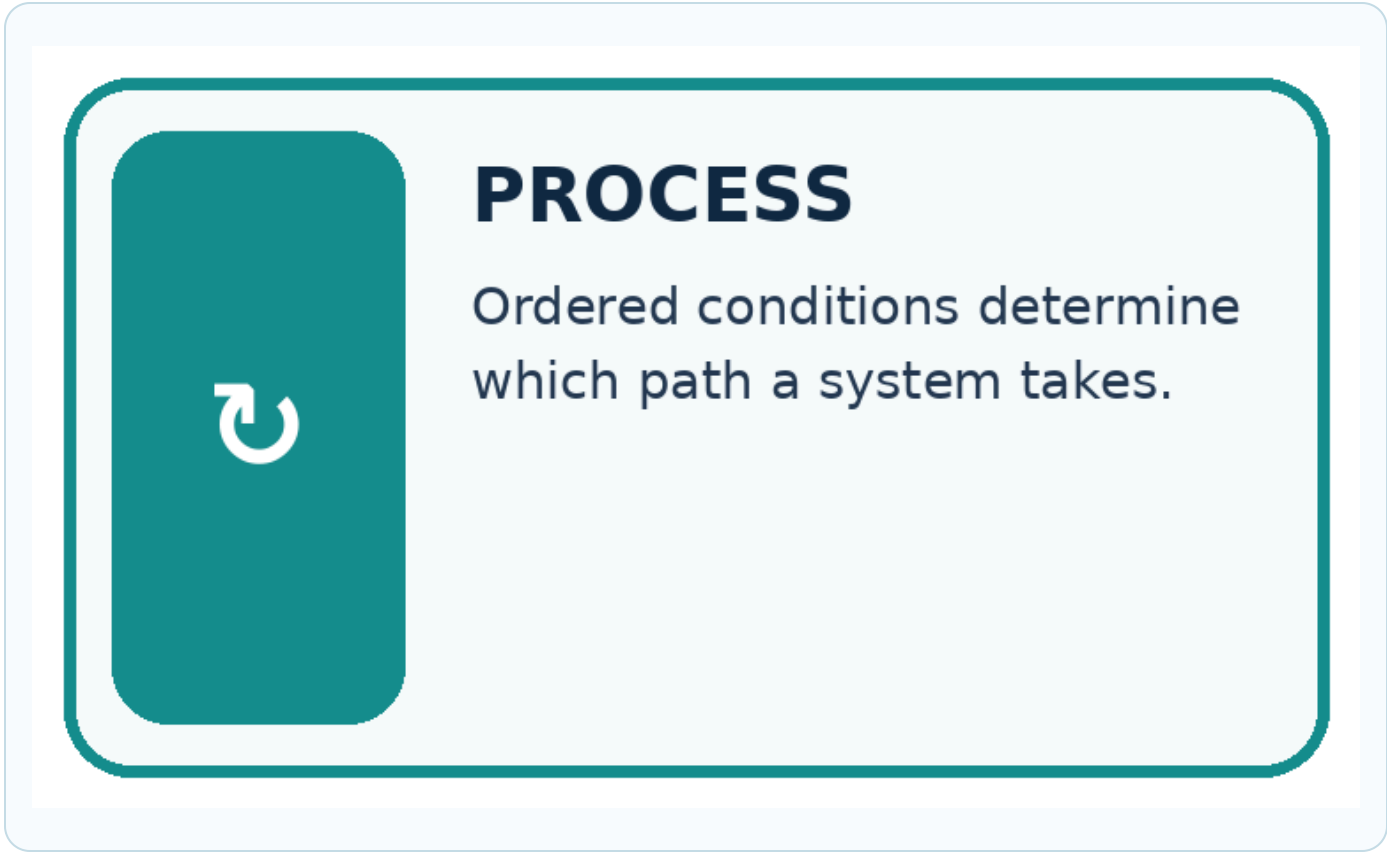
VISUAL MECHANISM

Input → Process → Output

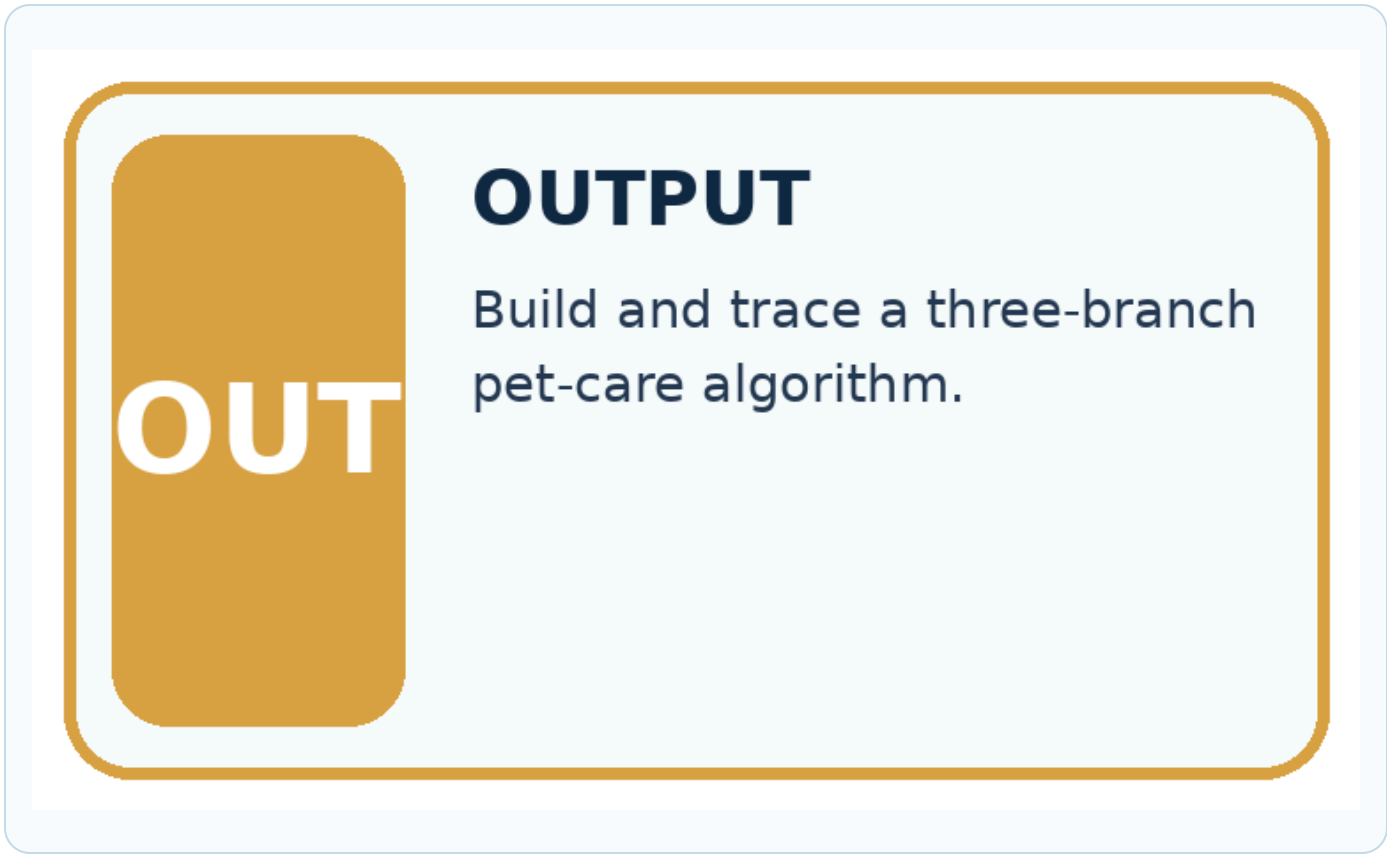
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Ordered conditions determine which path a system takes.

↓


OUT
OUTPUT
Build and trace a three-branch pet-care algorithm.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

One chain chooses one label

INPUT	CHECKS IN ORDER	SELECTED LABEL
-1	Below zero is True	Invalid
12	Below zero False; <= 12 True	Short
13	Both checks False	Long

Later branches in this chain are skipped after a condition succeeds.

```
minutes = 12
if minutes < 0:
    label = "Invalid"
elif minutes <= 12:
    label = "Short"
else:
    label = "Long"
print(label)
```

The **output** is Short. For 12, the first **condition** is false and the second is true. The else block is skipped. print(label) is not indented, so it runs after the selected **branch**.

Each path assigns label before this line reads it. Change only minutes and save before each fresh run.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

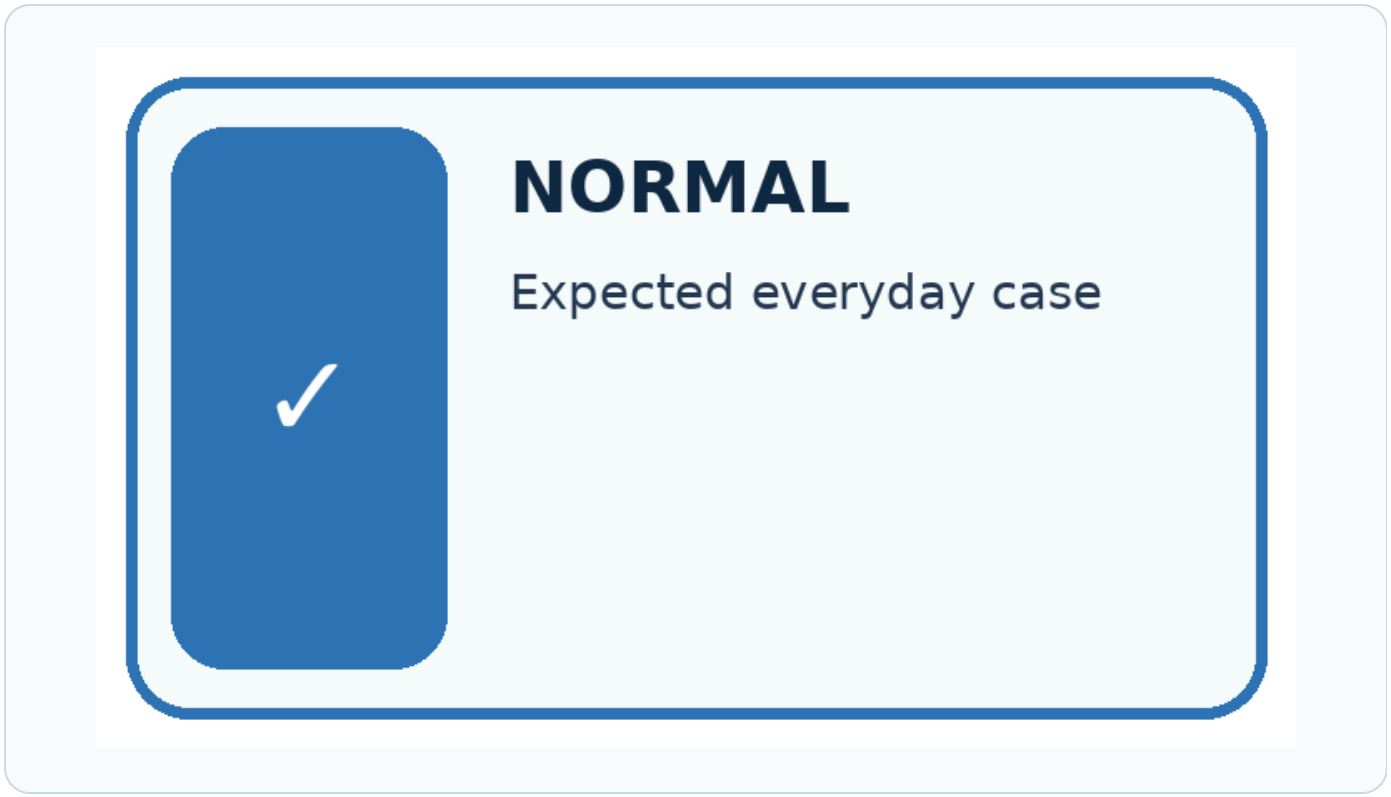
> Compare the supplied cases for Branching Logic. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

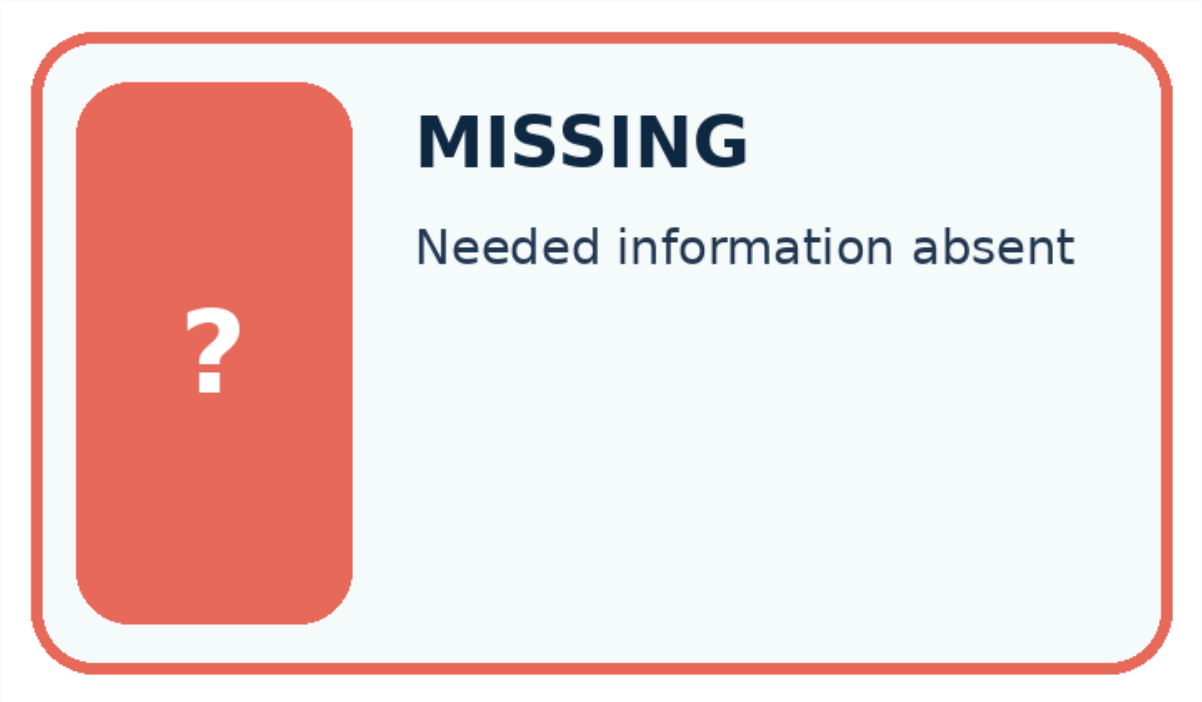
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. **BOUNDARY** CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Move minutes<=10 to the first **branch**.

Predicted result

Prediction: –1 and 0 wrongly receive Short.

Why the result follows

Reason: Both satisfy the earlier comparison before their intended checks are reached.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Branching Logic to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

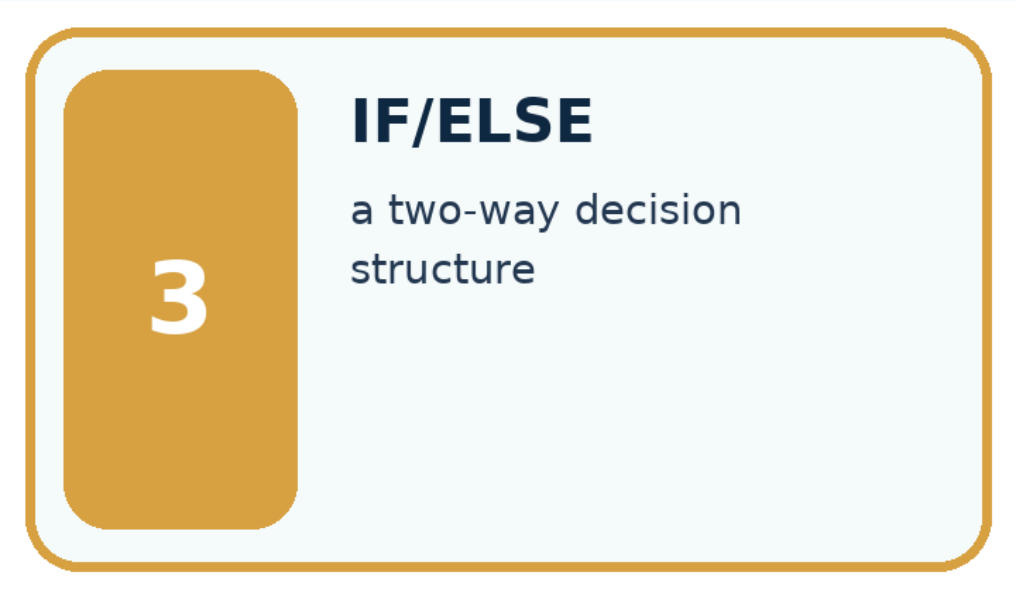
CONDITION

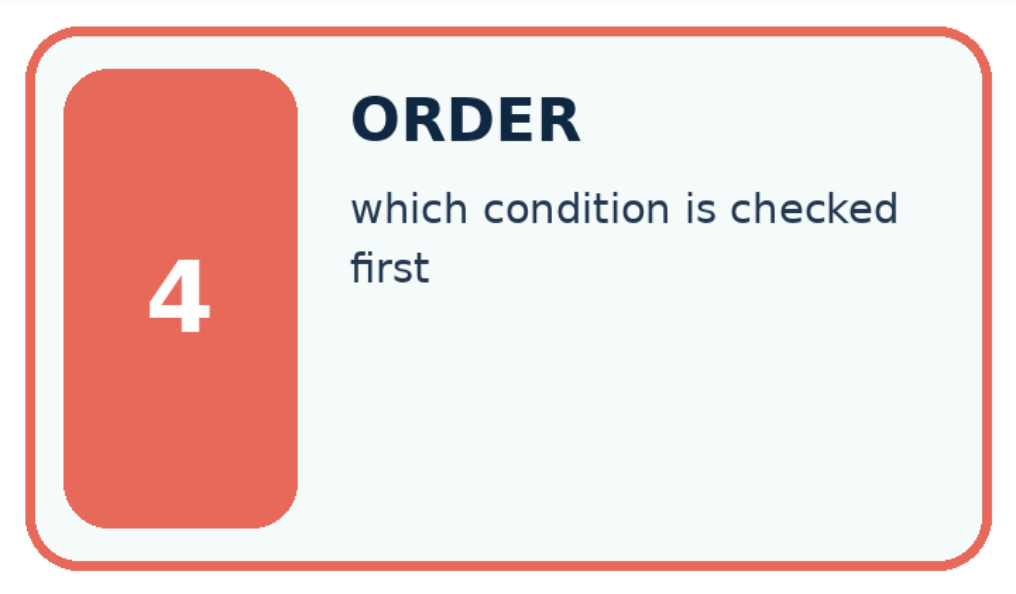
a question with a result

2

BRANCH

one path chosen after a condition





Word	My own example	Not this / boundary
condition	A statement evaluated as true or false to help select an action.	
branch	One path taken when a condition selects among alternatives.	
if/else	A programming structure that chooses a path according to whether a condition is true or false.	

Word	My own example	Not this / boundary
order	The sequence in which steps are performed or items are read.	

Words used in the closer explanation

Word	Meaning
Indentation	Leading spaces that identify a Python code block
Boundary	A value where a classification may change
Regression check	A previous test rerun after an edit to detect a new failure

---

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Branching Logic, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 3 • WORKED EVIDENCE

1. IF pet is hungry → feed.
2. ELSE IF pet needs exercise → walk or play.
3. ELSE → provide a quiet rest space.
4. Boundary test: what if hungry and needs exercise? The first true branch wins.

1. IF pet is hungry → feed.

2. ELSE IF pet needs exercise → walk or play.

3. ELSE → provide a quiet rest space.

4. **Boundary** test: what if hungry and needs exercise? The first true **branch** wins.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Input minutes	Expected label	Reason for this test
-1	Invalid	Negative input rejection
0	Short	Lowest valid duration
11	Short	Just below the upper boundary
12	Short	Exact upper boundary
13	Long	Just above the boundary

These tests exercise every **branch** and the equality case. A change from <= 12 to < 12 is exposed by 12; testing 11 alone misses it.

Branch coverage means every **branch** was exercised, but it does not prove every **condition** is correct.

Suppose the program uses < 12 instead of <= 12. What is printed for 12, why does that violate the written rule, and which earlier test should still pass after repair?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Branching Logic

A fictional reading-time labeler uses one Python if/el**if/else** chain.

The facts we will use

> **Supplied facts**
>
> Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long.

> **Supplied facts**
>
> **Order** checks as minutes<0, minutes==0, minutes<=10, otherwise. Print the assigned label once.

AI ACADEMY / AI-2/3

Branching Logic

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Negative check

-1 takes Invalid and skips later branches.



2. Zero check

0 takes Empty, not Short.



3. Inclusive short range

1 and 10 take Short.



4. Remaining inputs

11 takes Long; only the selected branch assigns the label.

CHANGE ONE THING

Move minutes<=10 to the first branch.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Negative check	–1 takes Invalid and skips later branches.
2. Zero check	0 takes Empty, not Short.
3. Inclusive short range	1 and 10 take Short.
4. Remaining inputs	11 takes Long; only the selected branch assigns the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

What this example does not establish

Keep the stated limit

The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

Keep the stated limit

Read the labelled picture: Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short.

Keep the stated limit

Change one thing: Move minutes<=10 to the first **branch**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Branching Logic claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: All true conditions run even in an if/else-if chain. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

For each run, record **input**, expected label, **condition**s actually checked and observed label. Mark skipped **condition**s as skipped rather than pretending the program evaluated them.

After a repair, rerun earlier cases as well as the failing one. Week 6 will repeat a block and track changes across iterations.

A learner tests -1, 2 and 20 and concludes the sorter is completely correct because all three branches ran. Explain what important case is missing and why branch coverage alone is insufficient.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Branching Logic with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one **boundary** case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

```
minutes = -1
if minutes < 0:
    label = "Invalid"
if minutes <= 12:
    label = "Short"
else:
    label = "Long"
print(label)
```

This faulty version prints Short. The second if starts a separate decision and overwrites Invalid. Its else belongs to that second if. Restoring elif makes the second test depend on the first test failing.

Moving print inside one **branch** can also hide **output** on other paths; **indentation** affects behavior, not just appearance.

The script rejects negative integers, but it does not validate arbitrary types or check a measured duration. A supplied text value can raise an error during comparison.

Keep the stated integer **input** contract and report the limitation honestly.

Practise with less help: Trace the tested path

Use workbook W2 for the connected example "**Branch**ing and **boundary** tests". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Branch**ing and **boundary** tests. Proposed explanation: "Every **condition** in the chain always runs". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

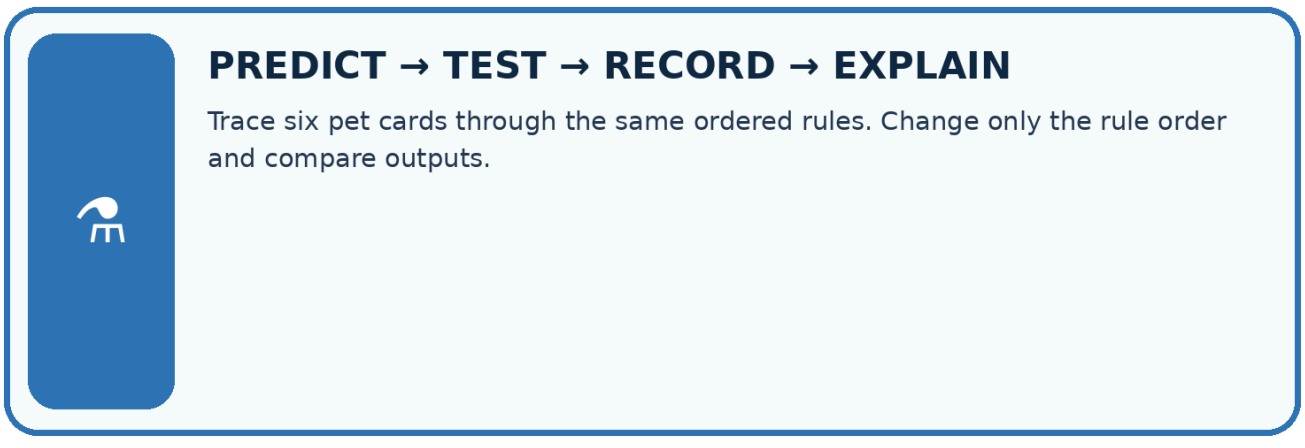
Run the investigation

> Investigate Branching Logic. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Trace six pet cards through the same **order**ed rules. Change only the rule **order** and compare **output**s.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

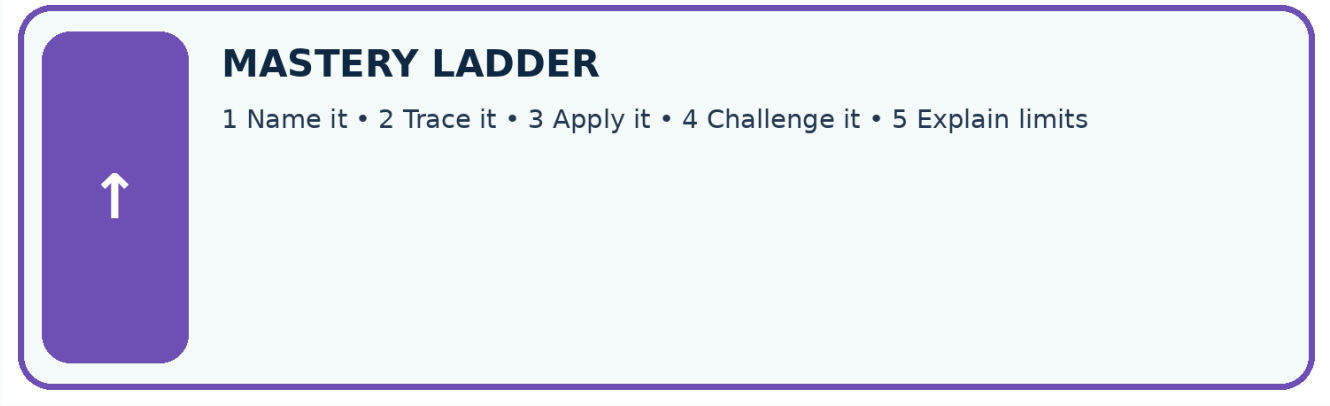
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Branching Logic. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, **branch**, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a **boundary** case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT **Branch** when a reader has a hard constraint such as language, format, or maximum length.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

STUDENT LESSON · 12 OF 18

Show what you know

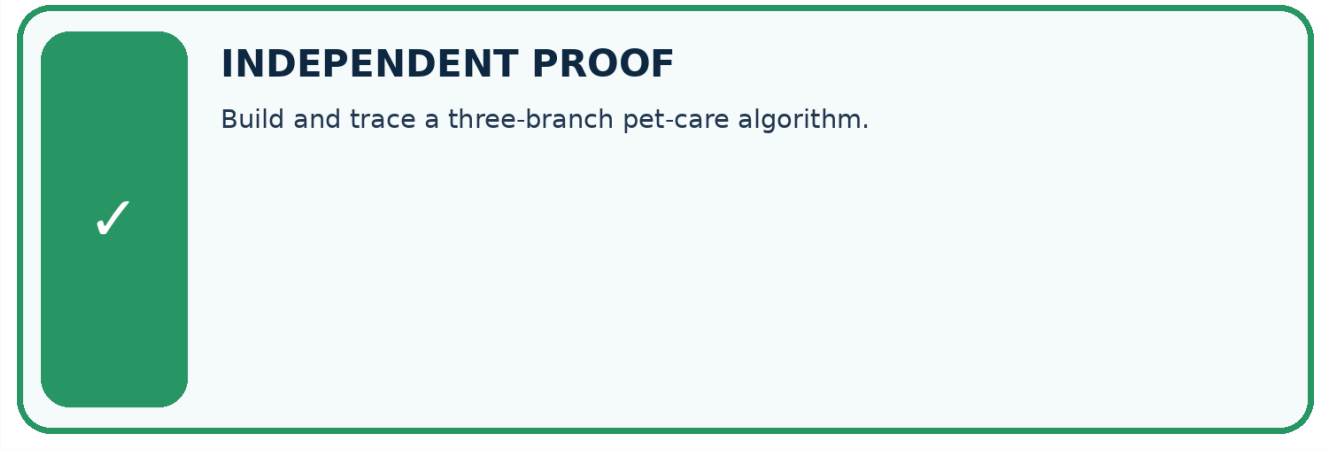
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Build and trace a three-**branch** pet-care algorithm.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a **boundary** or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Branching Logic case for supported practice. Repeated examples do not become fresh assessments.

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

The question

Classify 99,100,200,201 pages.

Worked explanation

Short, Medium, Medium, Long. Strict and inclusive boundaries and **branch order** matter.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Test strict and inclusive page boundaries

Evidence or stage	Value or result
99 pages	Less than 100 → Short
100 pages	100 to 200 inclusive → Medium
200 pages	100 to 200 inclusive → Medium
201 pages	Greater than 200 → Long

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Short, Medium, Medium, Long. Strict and inclusive boundaries and **branch order** matter.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an **order**ed if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

If pages are at most 80, choose short; otherwise choose long. Classify books with 79, 80 and 81 pages and explain the **boundary**.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a **boundary** or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Branch when a reader has a hard constraint such as language, format, or maximum length.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 **branch** is placed first?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Branching Logic, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional reading-time labeler uses one Python if/el**if/else** chain.

By the end, I can	How I will show it
Trace the selected branch in an if elif else chain	Explain the deciding step, show a trace or test, and state one limit.
Use indentation to identify which statements belong to a branch	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Test values at and around a decision boundary	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Branching Logic without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace the selected **branch** in an if elif else chain

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-time labeler uses one Python if/el**if/else** chain.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 11 takes Long; only the selected **branch** assigns the label.

1. Negative check

2. Zero check

3. Inclusive short range

4. Remaining **input**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Remaining **input**s 11 takes Long; only the selected **branch** assigns the label.

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move minutes<=10 to the first **branch**. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. –1 and 0 wrongly receive Short.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: –1 and 0 wrongly receive Short. Both satisfy the earlier comparison before their intended checks are reached.

Repair a missing building block

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

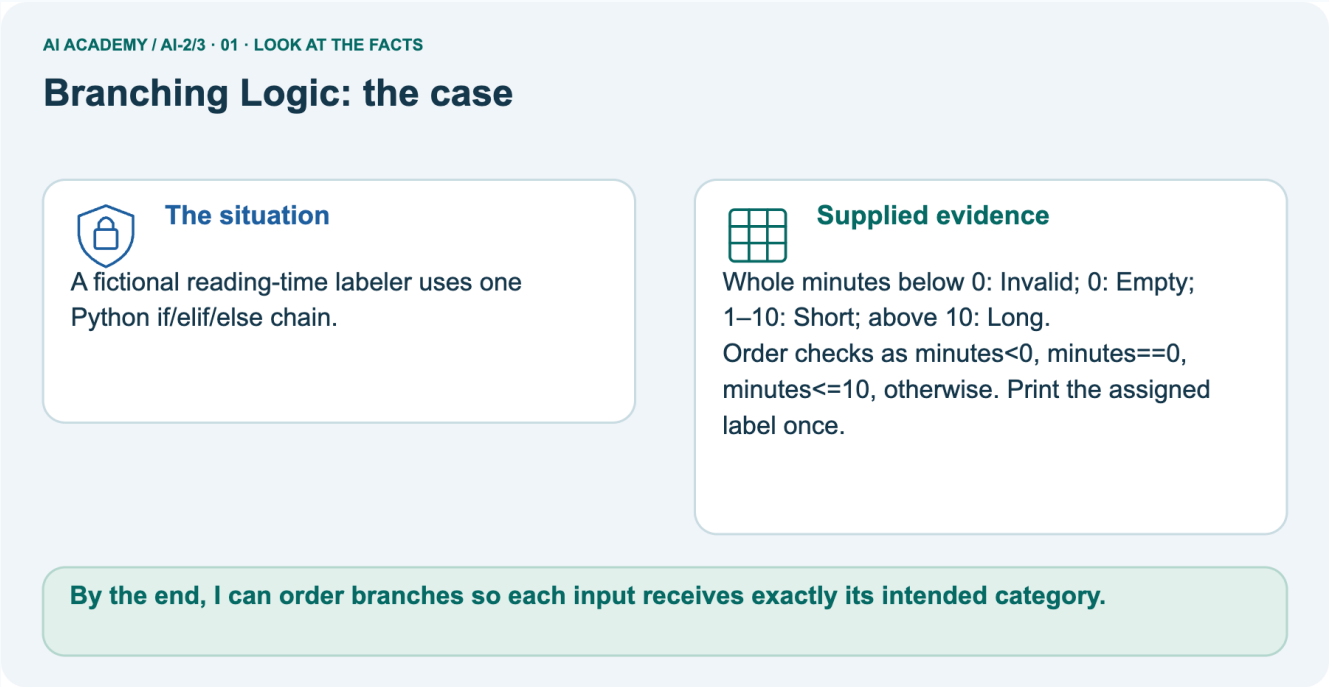
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Branching Logic graphic using its labels, deciding evidence and relevant limitation.

01. Branching Logic: the case



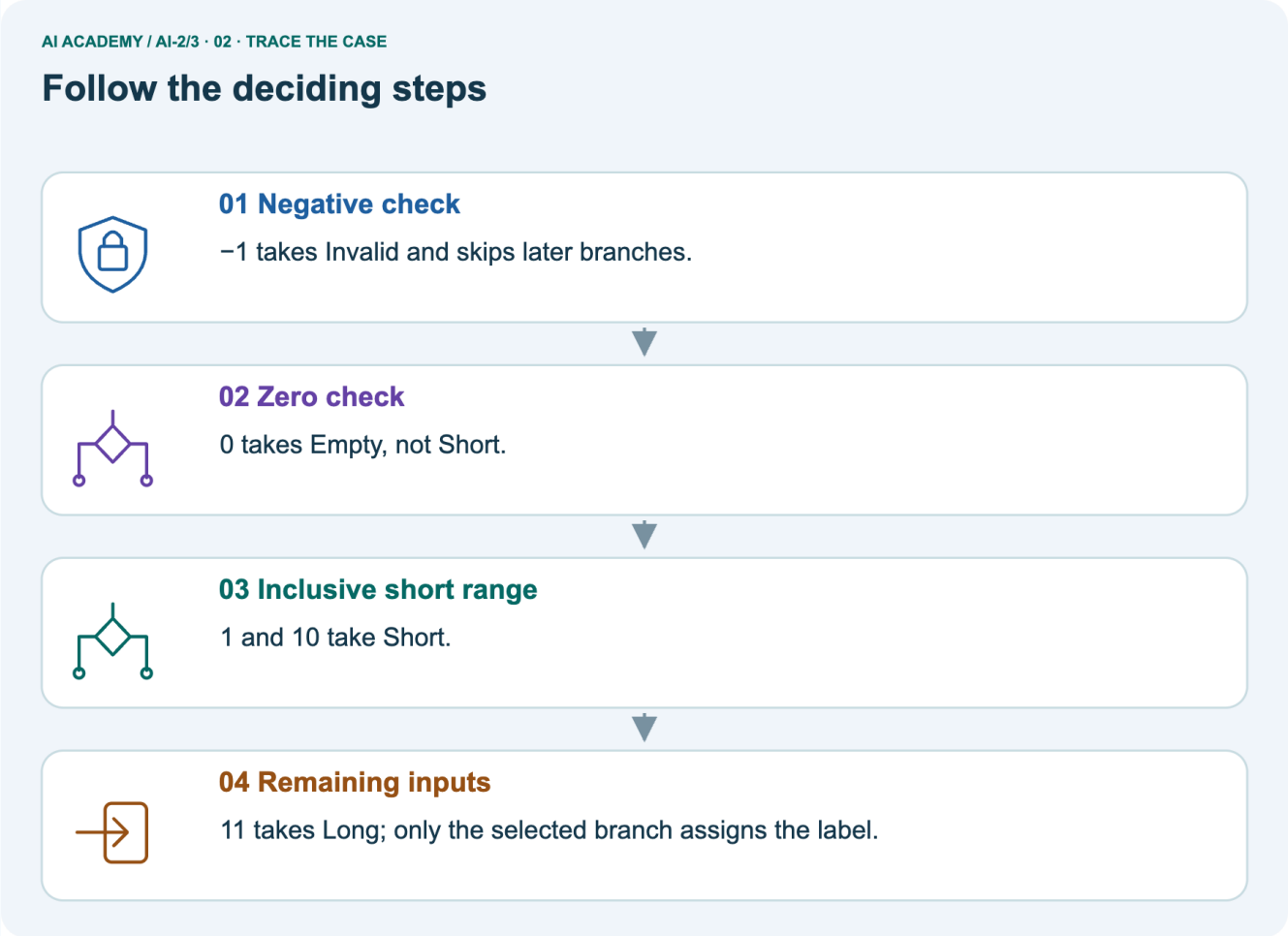
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional reading-time labeler uses one Python if/el**if/else** chain. Supplied facts: Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long. **Order** checks as minutes<0, minutes==0, minutes<=10, otherwise.

Print the assigned label once.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Negative check: –1 takes Invalid and skips later **branch**es. → Zero check: 0 takes Empty, not Short. → Inclusive short range: 1 and 10 take Short. → Remaining **input**s: 11 takes Long; only the selected **branch** assigns the label.

Point to each numbered stage and explain which recorded **input** it uses.

03. One chain chooses one label

AI ACADEMY / AI-2/3 · 03 · INSPECT THE LESSON MODEL

One chain chooses one label

Input	Checks in order	Selected label
-1	Below zero is True	Invalid
12	Below zero False; <= 12 True	Short
13	Both checks False	Long

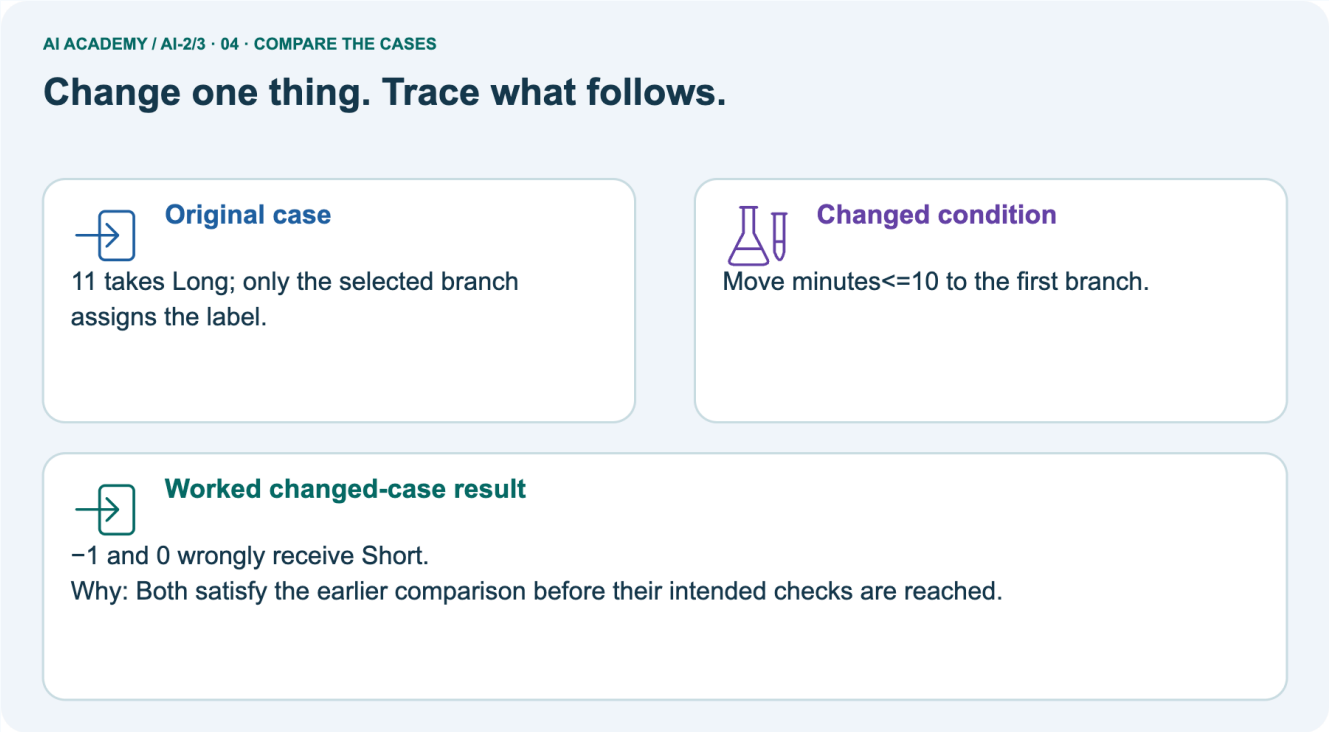
Later branches in this chain are skipped after a condition succeeds.

Follow the labelled picture with the matching explanation. Open the vector diagram

One chain chooses one label. -1 | Below zero is True | Invalid. 12 | Below zero False; <= 12 True | Short. 13 | Both checks False | Long

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



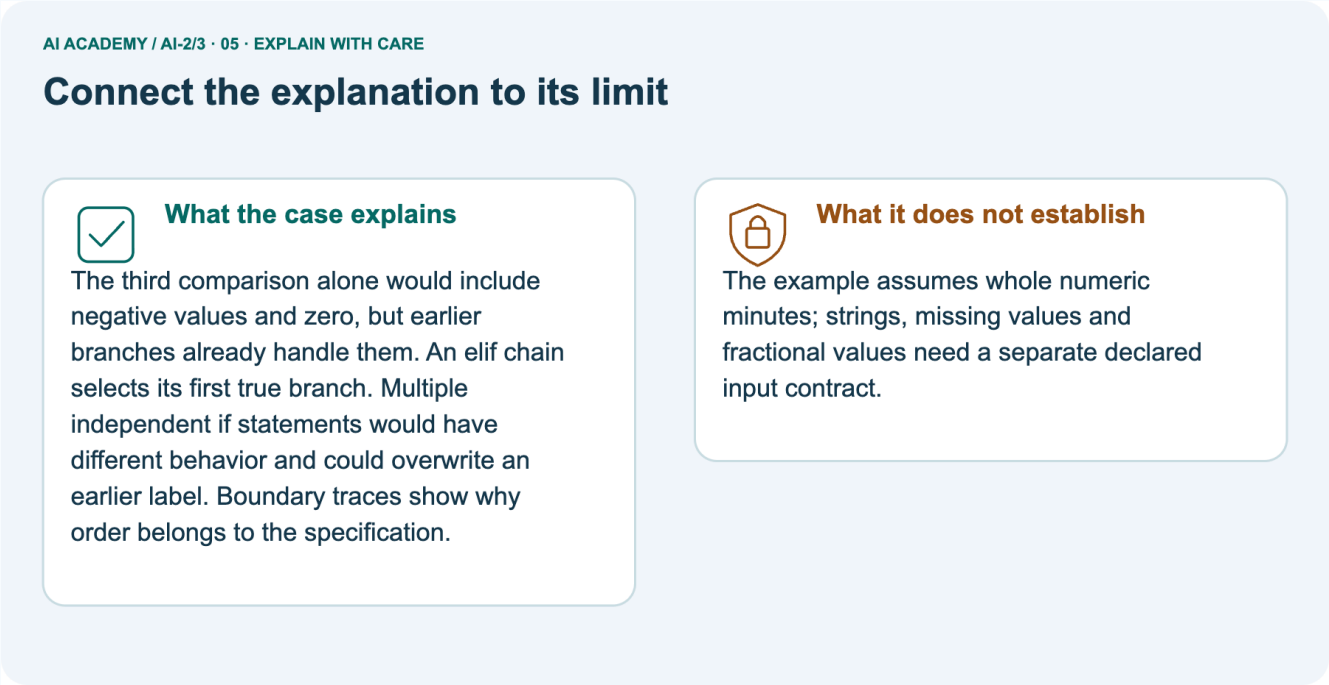
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 11 takes Long; only the selected **branch** assigns the label. Changed **condition**: Move minutes<=10 to the first **branch**. Expected result: −1 and 0 wrongly receive Short.

Reason: Both satisfy the earlier comparison before their intended checks are reached.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

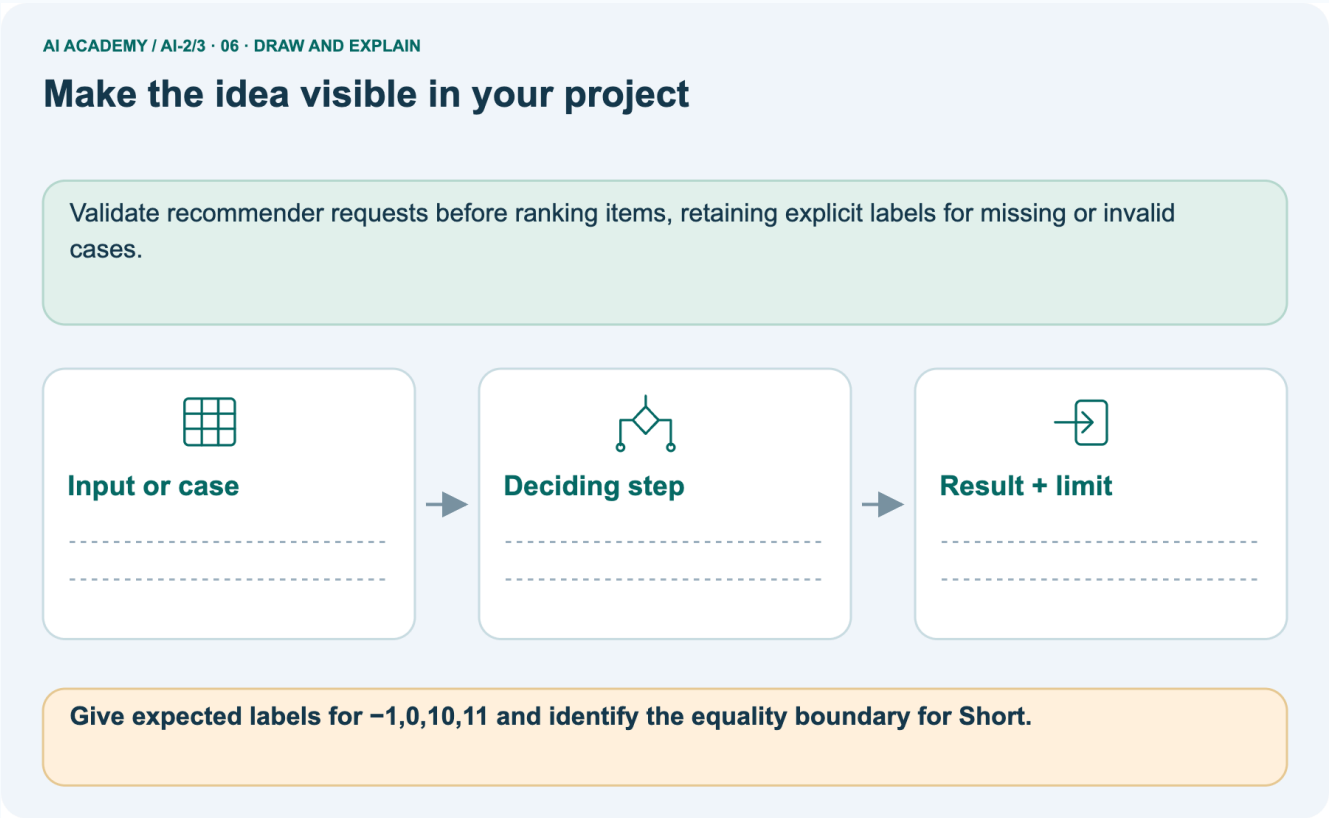
The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Limit: The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases. Draw your **input**, deciding step, result and limit.

Supported question: Give expected labels for −1,0,10,11 and identify the equality **boundary** for Short.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.